Background:

Article#: BIOFLEX-LAB107AW, Material composition is 84% Recycled Nylon, 16% Spandex with Wicking treatment, there are totally 4 colors, but only this particular RUM colored pants had this color shading issue

Phenomenon:

Color shading in Finished Pants: Minor VS. Severe/Worst -




VS.


Description:

RUM 色褲子出現明顯色差, 照片中標記為 “Worst” 的樣品可見局部顏色深淺不一, 尤其在袋口, 縫線及部分布面有明顯色塊, 屬於嚴重不均勻染色現象.

The RUM-colored pants showed obvious color differences, and the samples marked as "Worst" in the photo showed different shades of color, especially on the bag opening, stitches and some cloth surfaces, which has serious uneven dyeing phenomenon.

標記為 “Minor” 的樣品則顏色較為均勻, 僅有輕微色澤變化, 屬於可接受範圍.

Samples labeled "Minor" is more uniform in color, with only a slight change in color, which is within Acceptable Quality Level.

可能造成原因/Possible Root Cause:

酸性染料與吸濕排汗劑的化學反應: RUM 色專用的酸性染料與吸濕排汗劑之間可能產生化學不相容, 導致染料在尼龍纖維上的固著不均, 造成色差.

Chemical reaction between acid dyes and moisture-wicking agents: Chemical incompatibility may occur between acidic dyes and moisture-wicking agents dedicated to RUM color, resulting in uneven fixation of dyes on nylon fibers, resulting in color difference.

僅 RUM 色受影響, 其他三色未出現此問題, 顯示是特定染料配方與後整理劑/吸濕排汗劑之間的專一性反應.

Only the RUM color was affected, and the other three colors did not have this problem, indicating a specific reaction between the specific dye formulation and the finisher/moisture-wicking agent.

POV:

RUM 色褲子的色差問題, 極有可能來自酸性染料與吸濕排汗劑的化學不相容, 以及尼龍/彈性纖維混紡材料對染色工藝的敏感性, 建議從染料選擇, 工藝流程及後整理劑的相容性三方面進行優化.

The color difference of RUM pants is most likely due to the chemical incompatibility between acid dyes and moisture-wicking agents, as well as the sensitivity of nylon/elastin blended materials to the dyeing process.

這種色差問題是極有可能在布料階段沒發現, 到了成衣廠才顯現的, 原因如下:

This color difference problem is very likely NOT to be detected at the fabric stage, and only appears in the garment factory for the following reasons:

布料與成衣加工流程差異/ Different process in between fabrics and garments

布料在染整廠時通常是大面積, 平整狀態進行檢查, 表面光線均勻, 且未經縫製, 壓燙, 拉伸等後續加工. The fabric is usually inspected in a large area, flat state, uniform surface light, and has not been sewn, pressed, stretched and other subsequent processing.

成衣廠在裁剪, 縫製, 整燙過程中, 布料會經過多次彎折, 拉扯, 壓力及熱處理, 這些都可能影響表面顏色的呈現, 尤其是對染色均勻性敏感的布種. In the process of cutting, sewing and ironing, the fabric will be bent, pulled, pressed and heat treated many times, which may affect the presentation of surface color, especially for fabrics that are sensitive to dyeing uniformity.

成衣局部加工或熱處理/Garment with semi-assembly or heat treatment

成衣製作時局部（如袋口、縫線、腰頭）會有較高的壓力與熱度, 這會讓原本微小的色差被放大, 甚至出現明顯的色塊或陰影. During the production of garments, there will be high pressure and heat in some parts (such as bag openings, stitches, waistbands), which will magnify the original small color differences, and even obvious color blocks or shadows.

布料檢驗方式有限/Limitation of Fabric Inspection/Testing

布廠檢驗多以目測為主, 且多數僅檢查表面明顯瑕疵, 對於潛在的化學相容性問題或後續加工反應較難發現. Most of the garment factory inspections are mainly focus on visual inspections, and most of them only check the obvious defects on the surface, which is difficult to find potential chemical compatibility problems or subsequent processing reactions.

有些色差只有在特定角度, 光源或成衣狀態下才會明顯顯現. Some color differences are only noticeable at certain angles, light sources or garments.

Example 1. 某些尼龍/彈性布料在布廠檢驗時看起來均勻, 但成衣廠高溫壓燙後, 局部出現顏色深淺不一. Some nylon/elastic fabrics look uniform during inspection in the fabric factory, but after the garment factory is pressed at high temperature, there are different shades of color in some parts.

Example 2. 吸濕排汗劑在布廠與染料無明顯反應, 但成衣廠再次加熱定型後, 產生化學變化, 導致顏色變淡或產生色花. The hygroscopic wicking agent does not react significantly with the dye in the garment factory, but after the garment is heated and shaped again, it produces chemical changes, resulting in light color or color shading.